

Assignment 2

Due Date: Mar 23, 2026, at 11:59 PM

Notes:

- This is a **team assignment**. You will work together with your teammates to complete it.
- This assignment should be completed in either Java or Python.
- Zip all your files (code files, README, test data, **team member information**, and any other necessary files) into a single archive named “CPSC371_a#_teamname.zip” (or .rar, .7z, .tar). Submit this archive through Moodle.

Mushroom Classification

A mushroom, or toadstool is the fleshy, spore-bearing fruiting body of a fungus, typically produced above ground, on soil, or on its food source¹. In this assignment, we will work with mushrooms.

As we know, some mushrooms are edible, while others are poisonous. In this assignment, you are provided with a dataset file (MushroomData_8000.txt) which contains information on 8000 mushrooms. Your task is to build **two** different applications that can be used to predict whether an unknown mushroom is edible or poisonous. 100 unknown mushrooms are provided in a target dataset file (MushroomData_Unknown_100.txt) for you to run your application on. Detailed information about these datasets is described in later sections of this assignment.

Note: You will not be provided with the correct answers for these unknown mushrooms. I will evaluate the accuracy of your program during grading. However, only a portion of the marks will depend on the accuracy of your application, so don't worry too much about achieving perfect accuracy.

Applications and Algorithms

You will use the following two techniques to implement **two separate applications**:

1. KNN, and
2. Perceptron

For both applications, you have the freedom to implement them any way you choose:

- You can divide the training and testing dataset in any way you like (80-20, 50-50, etc.).
- You can set K to any value you feel appropriate in KNN.
- You can design and train your perceptron with any settings you like (λ , γ , starting values for the training weight set, etc.).
- You can preprocess the dataset in any way you like.
- And so on.

¹ <https://en.wikipedia.org/wiki/Mushroom>

Dataset Information

In **MushroomData 8000.txt**, the data is organized as follows:

- Each row contains a set of information for one mushroom (8000 mushrooms in total).
- Each row contains a total of 23 attributes for each mushroom:
 1. **classes:** **edible=e, poisonous=p**
 2. cap-shape: bell=b,conical=c,convex=x,flat=f,knobbed=k,sunken=s
 3. cap-surface: fibrous=f,grooves=g,scaly=y,smooth=s
 4. cap-color: brown=n,buff=b,cinnamon=c,gray=g,green=r,
pink=p,purple=u,red=e,white=w,yellow=y
 5. bruises?: bruises=t,no=f
 6. odor: almond=a,anise=l,creosote=c,fishy=y,foul=f,
musty=m,none=n,pungent=p,spicy=s
 7. gill-attachment: attached=a,descending=d,free=f,notched=n
 8. gill-spacing: close=c,crowded=w,distant=d
 9. gill-size: broad=b,narrow=n
 10. gill-color: black=k,brown=n,buff=b,chocolate=h,gray=g,
green=r,orange=o,pink=p,purple=u,red=e,
white=w,yellow=y
 11. stalk-shape: enlarging=e,tapering=t
 12. stalk-root: bulbous=b,club=c,cup=u,equal=e,
rhizomorphs=z,rooted=r,missing=?
 13. stalk-surface-above-ring: fibrous=f,scaly=y,silky=k,smooth=s
 14. stalk-surface-below-ring: fibrous=f,scaly=y,silky=k,smooth=s
 15. stalk-color-above-ring: brown=n,buff=b,cinnamon=c,gray=g,orange=o,
pink=p,red=e,white=w,yellow=y
 16. stalk-color-below-ring: brown=n,buff=b,cinnamon=c,gray=g,orange=o,
pink=p,red=e,white=w,yellow=y
 17. veil-type: partial=p,universal=u
 18. veil-color: brown=n,orange=o,white=w,yellow=y
 19. ring-number: none=n,one=o,two=t
 20. ring-type: cobwebby=c,evanescent=e,flaring=f,large=l,
none=n,pendant=p,sheathing=s,zone=z
 21. spore-print-color: black=k,brown=n,buff=b,chocolate=h,green=r,
orange=o,purple=u,white=w,yellow=y
 22. population: abundant=a,clustered=c,numerous=n,
scattered=s,several=v,solitary=y
 23. habitat: grasses=g,leaves=l,meadows=m,paths=p,
urban=u,waste=w,woods=d

Notes: (1) As you can see, the first attribute (**1. classes**) is our target attribute. (2) Missing attribute values may appear in the dataset and are represented as “?”.

In **MushroomData Unknwon 100.txt**, every attribute except the class attribute is provided, since that is what your program must predict.

Application Output

All of your programs should output 1 file named `predictionResultXXX.txt` (XXX is KNN or PER), which contains the prediction results for all mushrooms in the unknown mushroom dataset `MushroomData_Unknwon_100.txt` in separate lines.

That is, your `predictionResultXXX.txt` should look like the following:

```
e
p
e
e
e
p
p
p
e
e
...
```

A Short Report

A short report is required for this assignment, in which the following information is required:

- How the data was divided (training/testing split)
- Final settings used for both applications
 - Example: value of K in KNN
 - Learning rate, number of iterations, or other parameters used in the Perceptron
- The accuracy on the test data for each application
- If multiple setups were tested, you may include a table or chart comparing different configurations
 - Example: different values of K in KNN.

Other Notes

1. For each technique, the application with the highest accuracy on the unknown data set (compared to all other students' work) will receive a small bonus mark (2%~5%).
 - a. If both of your applications have the highest accuracy, you will receive a $\times 2$ bonus.
2. **You are required to implement these applications from scratch. That means, you are NOT allowed to use any existing libraries that provide built-in functions of KNN or ANN.**
 - a. You are allowed to use libraries that provide basic functionalities (e.g., numpy, etc.).